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Asymmetric Simple Exclusion Process

This section follows notes on

Kardar-Parisi-Zhang and Stochastic Burgers

Kardar-Parisi-Zhang Modes in d-dimensional Directed Polymers

This section follows “Kardar-Parisi-Zhang modes in d-dimensional directed polymers”.

Example in d=3:

Take the start from a two-species asymmetric exclusion model on a ring with L sites where each site k is occupied by one of the three particle species written as X_k ,

$$A, \quad B, \quad \Phi$$

Here Φ represents vacancy. The Markovian stochastic dynamics consists of nearest neighbor particle exchanges, $X_i X_{i+1} \rightarrow X_{i+1} X_i$, with the following rates,

$$\begin{aligned} AB &\rightarrow BA \text{ with rate } r_1, & BA &\rightarrow AB \text{ with rate } r_2 \\ A\Phi &\rightarrow \Phi A \text{ with rate } r_3, & \Phi A &\rightarrow A\Phi \text{ with rate } r_4 \\ B\Phi &\rightarrow \Phi B \text{ with rate } r_5, & \Phi B &\rightarrow B\Phi \text{ with rate } r_6 \end{aligned}$$

Where,

$$r_1 + r_4 + r_5 = r_2 + r_3 + r_6,$$

to ensure equal equilibrium probabilities for all conformations of the polymer. Note this uniform distribution leads to a complete absence of stationary correlations in the thermodynamic limit $L \rightarrow \infty$.

The link with the height function and the two-dimensional random path is as follows; with each of the three species, A, B, Φ I was n